# rencing Report
# RRHÅLSKARTA

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Image file:  76D BORRHÅLSKARTA.jpg
Image size:  8544 × 5662 pixels
Number of GCPs:  16

Source CRS:  EPSG:3152  (ST74, Stockholm 1938)
Target CRS:  EPSG:3006  (SWEREF99 TM)

Transformation:  pyproj (PROJ pipeline)
Pipeline:  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808

Inter-consistency (affine fit):
  RMS residual:  1.293 m
  Max residual:  2.556 m
  Outlier threshold (2×RMS):  2.587 m

Overall quality:  GOOD
  (no outlier GCPs detected)
```

# GCP Locations — 76D BORRHÅLSKARTA

# Inter-consistency Analysis — Affine Fit Residuals

| GCP | Pixel | Line | ST74 X | ST74 Y | Res. X (m) | Res. Y (m) | Res. Total (m) | Flag |
|---|---|---|---|---|---|---|---|---|
| 1 | 1444 | 1205 | 100900 | 76400 | 0.067 | 0.575 | 0.579 | |
| 2 | 1480 | 2217 | 100900 | 76300 | -2.036 | 1.545 | 2.556 | |
| 3 | 1471 | 3209 | 100900 | 76200 | 0.616 | 0.543 | 0.821 | |
| 4 | 1487 | 4181 | 100900 | 76100 | 0.573 | -2.468 | 2.534 | |
| 5 | 3324 | 1210 | 101100 | 76400 | 0.030 | 0.084 | 0.089 | |
| 6 | 3337 | 2212 | 101100 | 76300 | 0.357 | 0.069 | 0.364 | |
| 7 | 3351 | 3213 | 101100 | 76200 | 0.577 | -0.047 | 0.578 | |
| 8 | 3364 | 4215 | 101100 | 76100 | 0.904 | -0.063 | 0.906 | |
| 9 | 5204 | 1215 | 101300 | 76400 | -0.008 | -0.406 | 0.406 | |
| 10 | 5217 | 2217 | 101300 | 76300 | 0.320 | -0.422 | 0.530 | |
| 11 | 5231 | 3218 | 101300 | 76200 | 0.539 | -0.538 | 0.762 | |
| 12 | 5274 | 4233 | 101300 | 76100 | -2.303 | 0.728 | 2.415 | |
| 13 | 7084 | 1220 | 101500 | 76400 | -0.045 | -0.897 | 0.898 | |
| 14 | 7097 | 2222 | 101500 | 76300 | 0.283 | -0.913 | 0.956 | |
| 15 | 7114 | 3249 | 101500 | 76200 | 0.227 | 1.564 | 1.581 | |
| 16 | 7133 | 4242 | 101500 | 76100 | -0.099 | 0.647 | 0.655 | |

```
Affine model:  map_x = 0.106407·pixel + -0.001707·line + 100748.34
               map_y = 0.000526·pixel + -0.099791·line + 76518.91

RMS residual: 1.293 m    |    Max residual: 2.556 m    |    Outlier threshold (2×RMS): 2.587 m

Grid spacing (map coords):  ΔX = [200.0, 200.0, 200.0]    ΔY = [100.0, 100.0, 100.0]
Pixel spacing (adjacent columns):  mean=1878.8  std=12.5  range=[1857, 1910]
Line spacing (adjacent rows):  mean=1001.8  std=12.8  range=[972, 1027]
```

## Coordinate Transformation — EPSG:3152 → EPSG:3006

| GCP | ST74 X | ST74 Y | SWEREF99TM E | SWEREF99TM N |
|---|---|---|---|---|
| 1 | 100900 | 76400 | 674783.757 | 6578450.491 |
| 2 | 100900 | 76300 | 674788.345 | 6578350.598 |
| 3 | 100900 | 76200 | 674792.934 | 6578250.706 |
| 4 | 100900 | 76100 | 674797.523 | 6578150.813 |
| 5 | 101100 | 76400 | 674983.542 | 6578459.669 |
| 6 | 101100 | 76300 | 674988.131 | 6578359.776 |
| 7 | 101100 | 76200 | 674992.719 | 6578259.883 |
| 8 | 101100 | 76100 | 674997.308 | 6578159.990 |
| 9 | 101300 | 76400 | 675183.328 | 6578468.846 |
| 10 | 101300 | 76300 | 675187.916 | 6578368.953 |
| 11 | 101300 | 76200 | 675192.505 | 6578269.061 |
| 12 | 101300 | 76100 | 675197.094 | 6578169.168 |
| 13 | 101500 | 76400 | 675383.113 | 6578478.024 |
| 14 | 101500 | 76300 | 675387.702 | 6578378.131 |
| 15 | 101500 | 76200 | 675392.291 | 6578278.238 |
| 16 | 101500 | 76100 | 675396.880 | 6578178.345 |

```
Source CRS:  EPSG:3152  —  ST74, Stockholm 1938 (local TM)
Target CRS:  EPSG:3006  —  SWEREF99 TM (ETRS89 / UTM zone 33N)

PROJ pipeline:
  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808 y_0=-6500614.7836 ellps=GRS80 step proj=utm zone=33 ellps=GR

This is equivalent to Lantmäteriet GTRANS.  The pipeline performs:
  1. Inverse ST74 Transverse Mercator projection (→ geographic on GRS80)
  2. Forward UTM zone 33 projection (→ SWEREF99TM)

No datum shift is required — both CRS use GRS80 ellipsoid and
SWEREF99/ETRS89 realization.  Round-trip accuracy: < 0.001 mm.
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